

A 1.06-to-5.09 TOPS/W Reconfigurable Hybrid-Neural-Network Processor for Deep Learning Applications

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Abstract

An energy-efficient hybrid neural network (NN) processor is implemented in a 65nm technology. It has two 16x16 reconfigurable heterogeneous processing elements (PEs) arrays. To accelerate a hybrid-NN, the PE array is designed to support on demand partitioning and reconfiguration for parallel processing different NNs. To improve energy efficiency, each PE supports bit-width adaptive computing to meet variant bit-width of different neural layers. Measurement results show that this processor achieves a peak 409.6GOPS running at 200MHz and at most 5.09TOPS/W energy efficiency. This processor outperforms the state-of-the-art up to 5.2X in energy efficiency.

Introduction

In practical deep learning applications, the underlying neural network is usually a hybrid of Convolutional Neural Network (CNN), Full Connection Network (FCN) and Recurrent Neural Network (RNN). For example, AlexNet is 5-layer CNN combined with 3-layer FCN. LRCN [1] for visual description is a hybrid of 5-layer CNN, 3-layer FCN and 2-layer RNN (Fig.1). State-of-the-art neural processors [2][3] are designed based on the homogenous MAC array considering CNN's data-reuse feature. When processing a hybrid-NN, FCN and RNN must be transformed as the expression of CNN and processed in a time multiplexing way [4]. This way of processing hybrid-NN has two drawbacks. First, since Computation to Communication Ratio of CNN, FCN and RNN are quite different, time multiplexing results in alternative under-utilization of either computing resources or memory bandwidth. Second, the weights in CNN, FCN and RNN can be quantized with different bit-width without accuracy loss. But the homogenous array cannot configure different bit-width for different MAC, resulting in low energy efficiency. In this work, we design a heterogeneous PE array based processor. In contrast with time multiplexing, it can be spatially partitioned on demand to process CNN, FCN and RNN in parallel. Moreover, it can be configured to construct variant bit-width data path. Therefore, it achieves high energy efficiency for processing arbitrary hybrid-NN.

Reconfigurable Heterogeneous Array based Architecture

Fig. 2 shows the top architecture of hybrid-NN Processor, including two 16x16 heterogeneous PE arrays, one controller, two 144KB multi-bank on-chip memory, a 1KB sharing weight buffer and a configurable interface. There are two types of PEs: general PE and super PE. General PE supports convolutional operation and full-connection operation. Super PE enhances general PE with five more operations: compare operation in pooling layer and tanh, sigmoid, multiplication, addition operations in RNN. As shown in Fig. 3, each PE is configured by a 12-bit configuration word (s0s1s2s3s4s5s6s7s8s9s10s11). Each array can be partitioned to at most 4 sub-arrays to process convolution, full-connection, pooling and RNN operations respectively. Three groups of I/O ports lie at the left, right and bottom of each array to guarantee independent I/Os for each sub-array. Adjacent PEs in each row are connected by two sets

of point-to-point data bus. Input data is transmitted to all PEs in one row of sub-array hop by hop, which enables data reuse in one row. The computation result is transmitted along opposite direction of input data. The PEs in one column share the same weights. Controller dynamically sends configuration words to each PE and sends weights to each PE column.

Bit-width Adaptive PE Computing

Each PE has two 8-bit multipliers. By separately controlling the carry bit of additions, each PE supports two computing modes. First, two 8-bit multipliers can be combined as a 16-bit multiplier to support 16-bit weights neural networks. Second, it can process two low bit-width (<8bit) networks in parallel. When the bit-width of weight in NN's layer is less than 8, two weights from different CNN kernels or FCN/RNN neurons are combined since they share the same input data. Therefore, each PE can complete two operations in parallel. The switching of computing mode is dynamically controlled by the s11 bit in configuration word to adaptively meet the bit-width of weights in each layer. It fully exploits the computing power of PE for low bit-width operations. Testing on low bit-width AlexNet (bit-width of 8 layers: 9/8/8/6/6/8/8/8) and LRCN (bit-width of 10 layers: 9/8/8/8/9/8/8/8/8/8), it improves energy efficiency by 1.06X and 1.17X compared with 16-bit AlexNet and LRCN.

On-demand PE Array Partitioning

There are two architectural features enabling on-demand PE array partitioning. First, each PE's function is independent configurable (Fig. 3) so that the PE array can be partitioned to different functional sub-array. Second, each PE can route data from/to adjacent PE and PE array has three groups of I/Os (Fig. 3), which guarantee each sub-array has its independent input/output path. For an arbitrary hybrid-NN, the PE array is partitioned for achieving highest performance according to NN's parameters, such as kernel size, the number of neurons, the number of iterations and *etc* (Fig. 4). Fig.5 shows the partitioning results of LRCN. The CNN sub-array is 13x9. The input data are propagated into array from left to right, and the computation results are propagated from right to left to go outside. 9 super PEs and 7 super PEs are assigned for pooling and RNN's operations respectively. The rest general PEs construct FCN sub-array. Contrary to CNN sub-array, the PEs in right most column load input data and propagate the input data from right to left, and the results are propagated from left. This on-demand array partitioning fully exploits the computing resources and bandwidth, it improves energy efficiency by 1.48X and 2.18X for AlexNet and LRCN respectively.

Measurement results

The test chip is fabricated in 65nm LPCMOS technology. Fig. 9 shows the die photo and chip spec. At room temperature, it runs at a nominal frequency of 200MHz at 1.2V, achieving 1.06TOPS/W efficiency. When scaling down from 1.2V to 0.67V, this chip achieves up to 5.09TOPS/W at 10MHz (Fig.6). Two hybrid-NN benchmarks: AlexNet and LRCN are run on the processor. When the processor runs at 16-bit time multiplexing mode, the energy efficiency is 0.8TOPS/W and

0.5TOPS/W respectively. If the on-demand array partitioning is adopted, it improves energy efficiency by 1.48X and 2.18X. Multiple sub-arrays enable parallel processing of different NNs, which resolves the under-utilization of hardware resource and increases efficiency. Therefore, the processor achieves throughput of 368.4GOPS and 381.2GOPS on AlexNet and LRCN. The bit-width adaptive computing can further improve energy efficiency by 1.06X and 1.17X (Fig.7). This processor achieves 1.27TOPS/W for 105fps AlexNet and 1.28TOPS/W for 83fps LRCN (227x227 ImageNet images) at 200MHz and 1.2V, outperforming the state-of-the-art up to 5.2X in terms of energy efficiency (Fig.8).

References

- [1] J. Donahue, et al., *WACV*, 2015
- [2] Y. H. Chen, et al., *ISSCC*, 2016.
- [3] B. Moons, et al., *VLSI Symp.*, 2016
- [4] Z. Chen, et al., *ICCAD*, 2016.

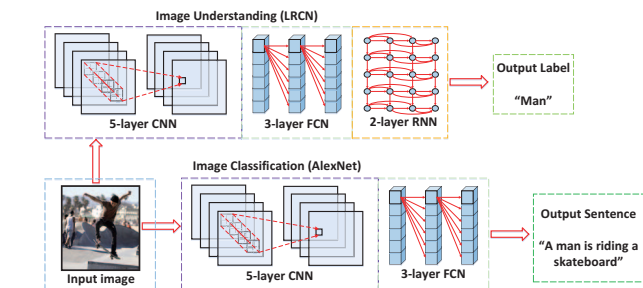


Fig.1 Hybrid neural networks in deep neural network

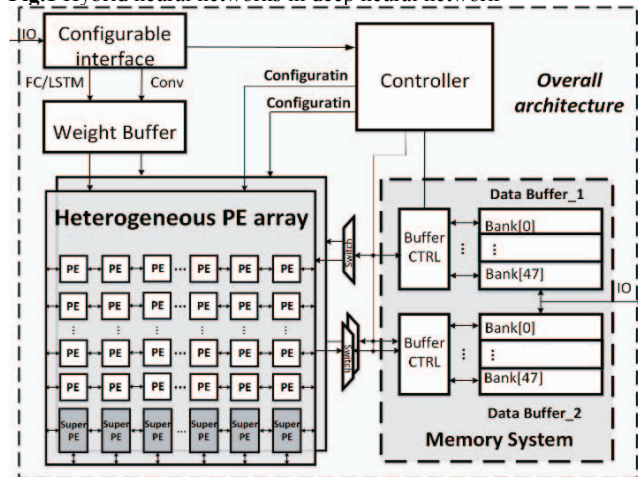


Fig.2 Top architecture of hybrid-NN processor

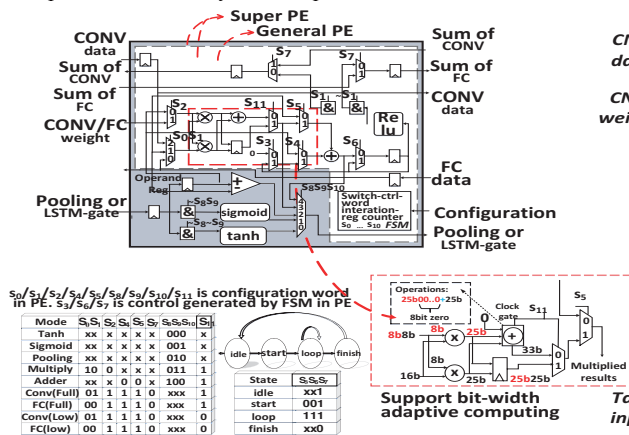


Fig. 3 PE architecture, configuration, data paths in PE for different functions.

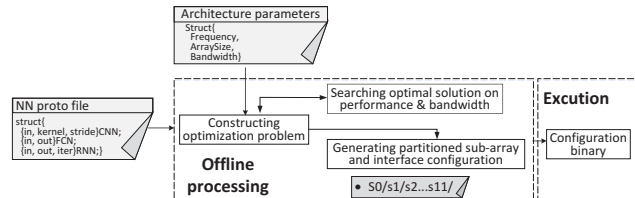


Fig. 4 Partitioning process

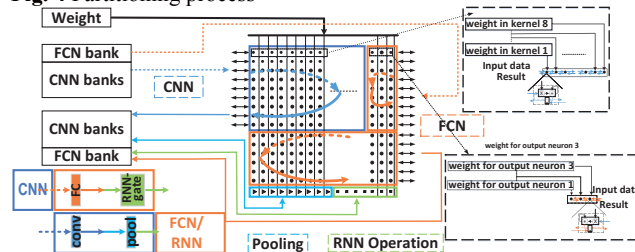


Fig. 5 Array partitioning and computation process of LRCN

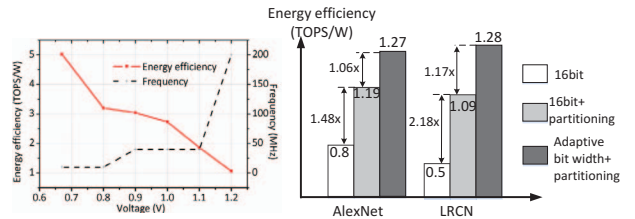


Fig.6 Efficiency on Voltage Scaling Fig.7 Performance of different strategy

	This Work	ISSCC'16 [2]	VLSI'16[3]
Frequency	200MHz	200MHz	204MHz
Area	4.4mmx4.4mm	3.5mmx3.5mm	1.2mmx2mm
On-chip SRAM	348KB	108KB	144KB
Technology	65nm/1.2 V	65nm/(0.82 ~ 1.17)V	40 nm/1.1V
Benchmark	AlexNet (CNN+FCN)	LRCN (CNN+FCN+LSTM)	AlexNet (Conv Only)
Throughput	368.4GOPS	381.2GOPS	67.6GOPS
Power	290mW	297.8mW	278mW
Energy efficiency	1.27TOPS/W	1.28TOPS/W	0.94TOPS/W

Fig. 8 Comparison with state-of-the-art.

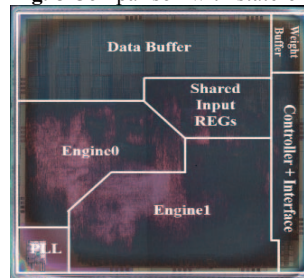


Fig. 9 Die photograph and Chip SPEC.

Technology	TSMC 65nm LP
Supply voltage	0.67V~1.2V
Area	4.4mm x 4.4mm
SRAM	348KB
Frequency	10~200MHz
Peak Performance	409.6GOPS
Power	4mW~447mW
Energy efficiency	1.06TOSP/W~5.09TOPS/W

